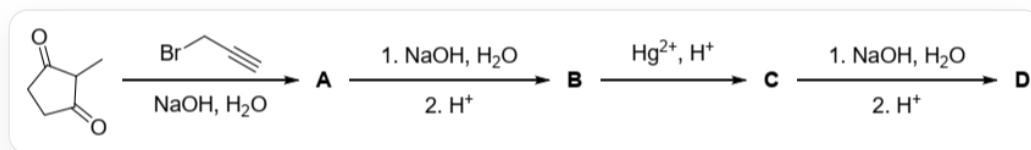


Question

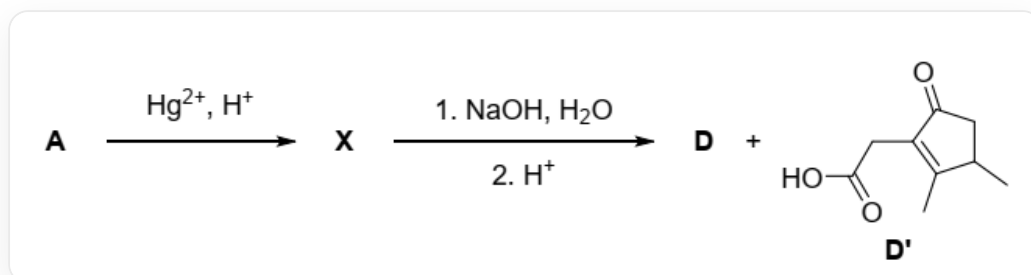
Compound **D** was obtained through the following reaction sequence.



The diagram contains the following reaction: CC1C(=O)CCC1=O>BrCC#C.[O-][Na+].O>[A], [A]>[O-].[Na+].O>[*], [*]>[H+]>[B], [B]>[Hg++].[H+]>[C], [C]>[O-].[Na+].O>[*], [*]>[H+]>[D], where the intermediate [*] is not shown in the original diagram

The chemical formula of **D** is $C_9H_{12}O_3$, and the molecule contains two methyl groups.

If the reaction from **A** to **B** is skipped and the process is directly carried out as follows, in addition to the main product **D**, a byproduct **D'** is also obtained.



[A]>[Hg++].[H+]>[X], [X]>[O-].[Na+].O>[*], [*]>[H+]>[D].[D'], where the structure of **D'** is CC1CC(=O)C(=C1C)CC(=O)O

Determine how many times ring-opening and ring-closing reactions occur in total during the reaction mechanism from **X** to **D'**.

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

G. 7

H. 8

I. 9

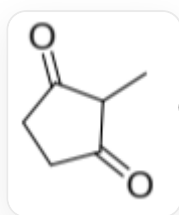
J. 10

Answer

Correct Answer: **F**

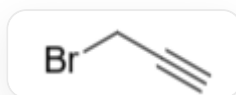
Detailed Explanation

The initial substrate is CC1C(=O)CCC1=O,



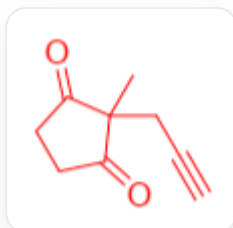
CC1C(=O)CCC1=O

in the presence of haloalkyne BrCC#C,



BrCC#C

and under basic aqueous solution conditions, alkylation occurs, and a propargyl group is attached to the common α -carbon of the two carbonyl groups, yielding **A**, C#CCC1(C)C(=O)CCC1=O.



C#CCC1(C)C(=O)CCC1=O

CHECKPOINT

1 PTS

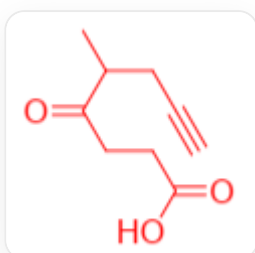
A propargyl group is attached to the common α position of the two carbonyl groups of the substrate

CHECKPOINT

1 PTS

A is C#CCC1(C)C(=O)CCC1=O

In an aqueous NaOH solution, the carbonyl carbon of **A** is attacked by OH^- , followed by ring-opening, breaking the bond between it and the common α -carbon of the diketone, and subsequent acidification yields the product **B** bearing a carboxyl group, C#CCC(C)C(=O)CCC(=O)O.



C#CCC(C)C(=O)CCC(=O)O

CHECKPOINT

1 PTS

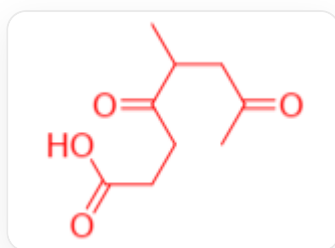
The carbonyl carbon of **A** is attacked by hydroxide and the ring opens, and acidification yields **B** bearing a carboxyl group

CHECKPOINT

1 PTS

B is C#CCC(C)C(=O)CCC(=O)O

From **B** to **C**, under acidic mercuric ion conditions, hydration of the terminal alkyne occurs, generating **C** with a methyl ketone structure, CC(CC(=O)C)C(=O)CCC(=O)O.



CC(CC(=O)C)C(=O)CCC(=O)O

CHECKPOINT

1 PTS

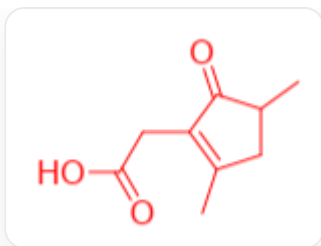
Hydration of the terminal alkyne of **B** yields **C** with a methyl ketone structure

CHECKPOINT

1 PTS

C is CC(CC(=O)C)C(=O)CCC(=O)O

The ketone carbonyl group near the end of the **C** molecule undergoes condensation with the α position of the central ketone carbonyl group under basic conditions, eliminating one molecule of water, generating the five-membered ring product **D**, CC1=C(CC(=O)O)C(=O)C(C)C1.



CC1=C(CC(=O)O)C(=O)C(C)C1

CHECKPOINT

1 PTS

C undergoes dehydration condensation to generate the five-membered ring product **D**

CHECKPOINT

1 PTS

D is CC1=C(CC(=O)O)C(=O)C(C)C1

The molecular formula of **D** is $C_9H_{12}O_3$. Compared with **A**, it has more C_3H_4O , which is equivalent to replacing one hydrogen atom with a propargyl group and adding one molecule of water, which is consistent with the above reasoning.

CHECKPOINT

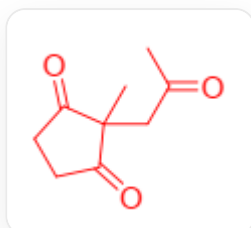
1 PTS

Compared with **A**, **D** is equivalent to replacing one hydrogen atom with a propargyl group and adding one molecule of water

It can be seen that there are indeed two methyl groups in the **D** molecule.

Next, analyze the process of $\mathbf{A} \rightarrow \mathbf{X} \rightarrow \mathbf{D} + \mathbf{D}'$.

From $\mathbf{A}$ to $\mathbf{X}$, again under acidic mercuric ion conditions, the terminal alkyne is oxidized to a methyl ketone, so the structure of $\mathbf{X}$ is CC(=O)CC1(C)C(=O)CCC1=O,



CC(=O)CC1(C)C(=O)CCC1=O

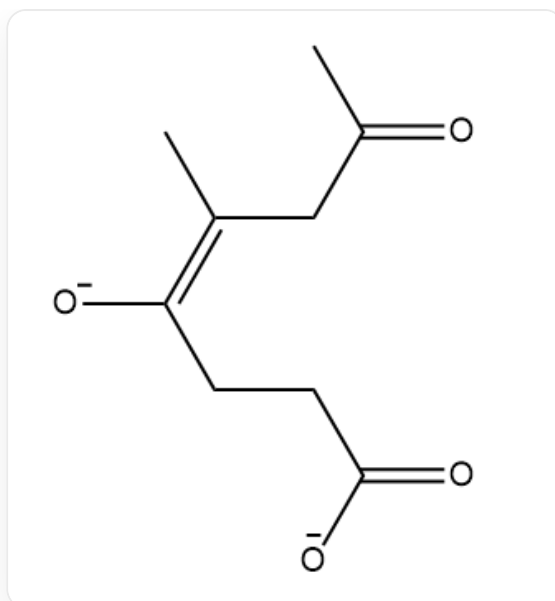
CHECKPOINT

1 PTS

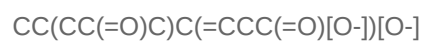
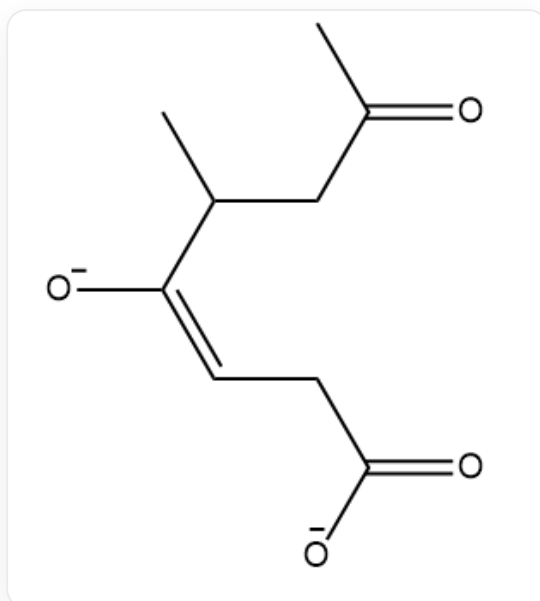
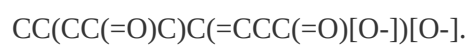
$\mathbf{X}$ is CC(=O)CC1(C)C(=O)CCC1=O

Under basic aqueous solution conditions, $\mathbf{X}$ is attacked by the nucleophile OH^- . Through a ring-opening - proton transfer - ring-closing process, $\mathbf{D}$ can also be obtained. The intermediates involved are

CC(=C(CCC(=O)[O-])[O-])CC(=O)C,



and

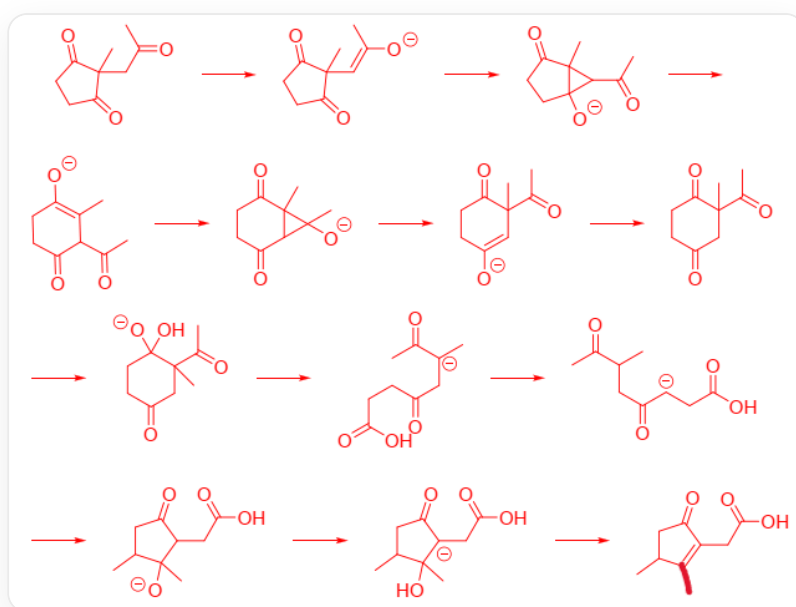


If OH^- does not act as a nucleophile to attack, but deprotonation occurs, then the subsequent reaction is:

CHECKPOINT

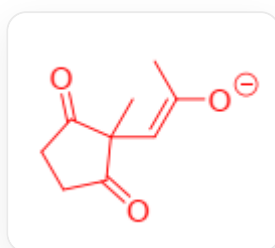
1 PTS

OH^- acts as a base to remove the α -position proton of the exocyclic carbonyl group in **X**



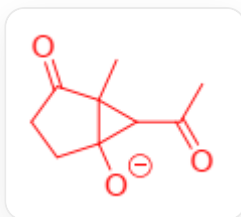
图中涉及13个分子或离子，通过箭头连接成一串，按顺序分别为CC(=O)CC1(C)C(=O)CCC1=O, C/C(=C\C1(C)C(=O)CCC1=O)/[O], CC(=O)C1C2(C)C(=O)CCC12[O], CC1=C(CCC(=O)C1C(=O)C)[O], CC12C(=O)CCC(=O)C1C2(C)[O], CC(=O)C1(C)C=C(CCC1=O)[O], CC(=O)C1(C)CC(=O)CCC1=O, CC(=O)C1(C)CC(=O)CCC1(O)[O], CC(=O)[C-](C)CC(=O)CCC(=O)O, CC(CC(=O)[C-]CC(=O)O)C(=O)C, CC1CC(=O)C(CC(=O)O)C1(C)[O], CC1CC(=O)[C-](CC(=O)O)C1(C)O, CC1CC(=O)C(=C1C)CC(=O)O

The specific process is that the α -position of the exocyclic carbonyl group of **X** is deprotonated to generate C/C(=C\C1(C)C(=O)CCC1=O)/[O]

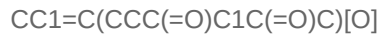
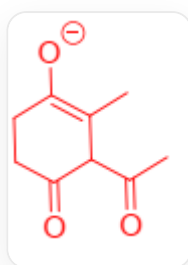


C/C(=C\C1(C)C(=O)CCC1=O)/[O]

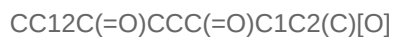
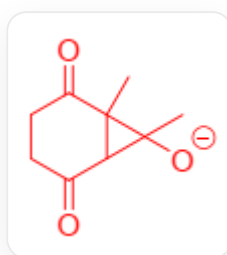
The enolate anion attacks the carbonyl carbon to obtain the [3.1.0] bridged ring CC(=O)C1C2(C)C(=O)CCC12[O]



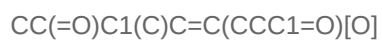
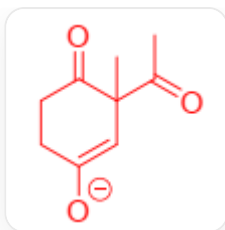
The carbon-carbon bond breaks to obtain an enolate anion with a six-membered ring structure CC1=C(CCC(=O)C1C(=O)C)[O]



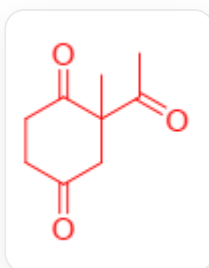
The enolate anion attacks the exocyclic carbonyl group to obtain the [4.1.0] bridged ring CC12C(=O)CCC(=O)C1C2(C)[O]



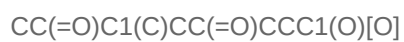
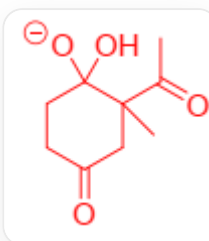
Ring opening yields another rearranged enolate anion CC(=O)C1(C)C=C(CCC1=O)[O]



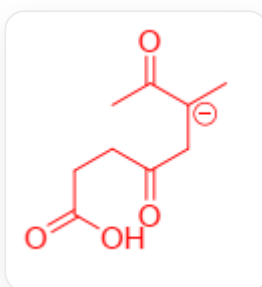
Protonation yields CC(=O)C1(C)CC(=O)CCC1=O



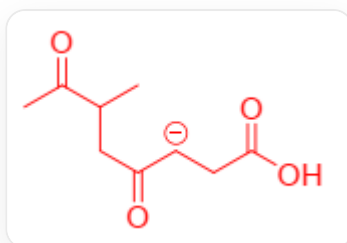
The carbonyl group is attacked by an external hydroxyl group CC(=O)C1(C)CC(=O)CCC1(O)[O-]



Ring opening yields the carboxylic acid intermediate CC(=O)[C-](C)CC(=O)CCC(=O)O

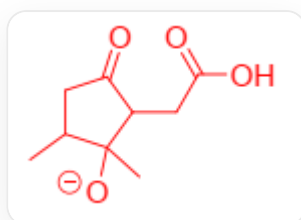


Proton transfer generates CC(CC(=O)[C-]CC(=O)O)C(=O)C



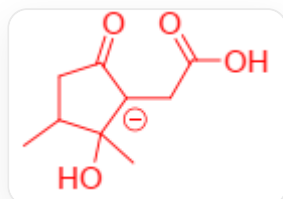
CC(CC(=O)[C-]CC(=O)O)C(=O)C

Ring closure generates the five-membered ring CC1CC(=O)C(CC(=O)O)C1(C)[O]



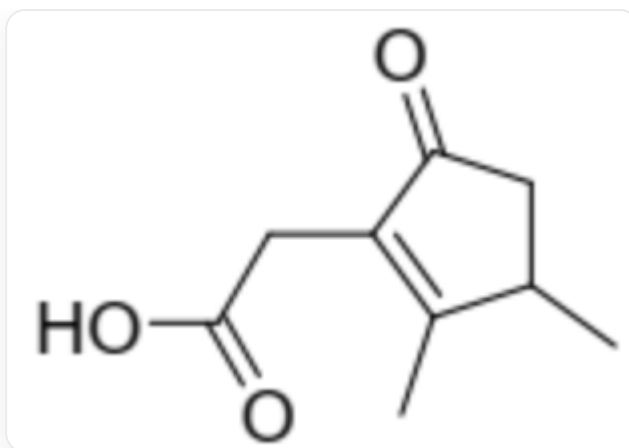
CC1CC(=O)C(CC(=O)O)C1(C)[O]

Proton transfer generates CC1CC(=O)[C-](CC(=O)O)C1(C)O



CC1CC(=O)[C-](CC(=O)O)C1(C)O

Hydroxyl leaves to generate **D'** CC1CC(=O)C(=C1C)CC(=O)O.



CC1CC(=O)C(=C1C)CC(=O)O

In the above process, a total of 3 ring closures and 3 ring openings occurred, so the total number of ring closures and ring openings involved is 6.

CHECKPOINT

1 PTS

3 ring closures, 3 ring openings